

Simant Singh

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Education

University of New Orleans Aug 2022 – May 2026
Bachelor of Science in Computer Science, Minor in Mathematics GPA: 3.9/4.0

Publications

- Raxit, S., **Singh, S.B.**, & Newaz, A.A.R. (2024). “YoloTag: Vision-based Robust UAV Navigation with Fiducial Markers.” *Proceedings of the 33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, DOI: 10.1109/RO-MAN60168.2024.10731319.
- Singh, S.B.**, Wagner, J., & Newaz, A. A. R. “Developing a Web-Based Dashboard for Multi-Modal Robotic Therapeutic Data” (Manuscript in preparation).

Experience

Bentley Systems Exton, PA (Remote)
Software Engineer Intern May 2025 – Aug 2025

- Led the development of an automated test framework in **Python**, improving regression testing efficiency by 76%.
- Optimized **PostgreSQL** queries and profiled memory leaks in a **Groovy** application to reduce production crashes by 27%.
- Delivered desktop-to-cloud sync using **Azure Service Bus** to ensure reliable data transfer in a hybrid cloud architecture.
- Collaborated in Agile sprints, conducted AI-assisted code reviews, and leveraged **CI/CD** for automated deployments.

Software Developer Intern May 2024 – Aug 2024

- Built and shipped client-facing full-stack features for a mobile app using **React Native** and **C#** for 5,000+ users.
- Engineered REST APIs and backend integrations using **.NET** and **SQL** to support a scalable microservices architecture.
- Designed reusable frontend UI components using **TypeScript** and **Tailwind CSS**, reducing development time by 30%.
- Implemented comprehensive unit and integration tests using **Jest** and **NUnit**, achieving 82% code coverage.

Autonomous Intelligent Robotics (aiR) Laboratory, University of New Orleans New Orleans, LA
Research Assistant Jan 2023 – May 2026

- Created **C++** computer vision components for YOLOTag using **OpenCV** and **PyTorch**, enabling real-time UAV navigation.
- Developed a full-stack web application using **React**, **Node.js**, and **Redis** for secure, real-time clinician data management.
- Designed scalable APIs using **FastAPI** and **MongoDB** to manage robotic system data across distributed systems.
- Built a backend integrating generative AI via **LLM APIs** and **MCP** to convert user prompts into actionable responses.
- Deployed an AI-driven research workflow system to **AWS** using **Docker** and **CI/CD** pipelines for a team of 14 researchers.
- Maintained technical documentation for APIs and system architectures, supporting long-term maintainability.

Projects

SensorySync (Startup): Co-founder and technical lead for a therapy coordination platform connecting parents and therapists with therapist-verified progress tracking, securing \$18,000 in seed funding to support technical development.

Transcript Slide Mapping: AI-driven NLP pipeline to align transcript lines with PowerPoint slides and a local distilled LLM to remove filler words and improve clarity. **Tools**: Python, TF-IDF, BERT, Sentence Transformers, Word2Vec, BM25

RESPER: Full-stack restaurant management application with admin, employee, and customer portals, deployed using AWS Lambda and S3. **Tools**: Vue, Express, MongoDB, AWS, Terraform, GitHub Actions

Car Dealership: Car dealing system with a chat application to buy and sell vehicles through agents. **Tools**: Spring Boot, Java, JUnit, TCP Socket, Object-Oriented Programming (OOP), Network Programming

Skills

Programming Languages: C, C++, C#, Java, Python, JavaScript, TypeScript, HTML/CSS
Frameworks & Libraries: React, Angular, Node.js, Express, .NET, Flask, FastAPI, PyTorch, TensorFlow
Databases & Storage: SQL (MySQL, PostgreSQL, Microsoft SQL Server), NoSQL (MongoDB, Redis)
Cloud & DevOps: AWS, Azure, Google Cloud, Docker, Terraform, CI/CD Pipelines (GitHub Actions)
Tools & Practices: Linux, Git, GitHub Copilot, API Design, Unit Testing, Code Reviews, Agile/Scrum